How a Simulator's Fidelity impacts the Performance of a Decision Support System: A Military Case Study

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***Abstract*—This study explores how simulator fidelity affects decision-support system (DSS) performance in a military mission scenario with real-world data. Seven quantifiable fidelity measures were introduced and tested. Experiments show that the "Building secured" and "maneuver" fidelity measures correlated positively with DSS accuracy. Findings highlight that fidelity measures can streamline DSS development, reducing costly re-training.**

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***Index Terms*—military, simulations, fidelity, decision support, machine learning**

I. Introduction

Decision-making is a critical aspect of Command and Control processes. As military operations increasingly span multiple domains and involve numerous dynamic components, a Decision Support System (DSS) may become indispensable for commanders. Recent advancements in computational power and artificial intelligence have led to a significant increase in the use of models and simulations as decision support tools in military operations. The primary goal of a DSS is to provide commanders and military personnel with relevant information, thereby enhancing the decision-making process and optimizing the tactical effectiveness of operations. Decision support applications can be employed for various purposes, including mission planning, human resource planning, and logistics planning [1], [2].

For decision making on the tactical level, NATO uses the APP-28 process [3], which shares the same characteristics and is very similar to the Military Decision-Making Process (MDMP) as described by the United States Army Doctrine Publication 5-0 The Operations Process [4]. During this process, a Course of Action (COA) is drafted and compared to the alternatives. By creating a simulation model, it becomes possible to generate synthetic data, develop COAs, test their feasibility, evaluate their effectiveness, and compare them. The outcomes of these simulations primarily serve as a support tool, offering valuable insights and recommendations to the commander by providing estimates of mission duration, potential casualties, and material damages. DSSs can adjust mission parameters, simulate each parameter set, and assess the effectiveness of achieving the commander's objectives based on the simulation results [5]. Given the complexity of COAs, we find that the simulation results are particularly well-suited to support specific sub-aspects or roles in the planning process, such as those of the military engineer [6].

DSSs typically consists of three parts: the model, the database, and the human interface [7]. In terms of the database, the question that is being studied in this article is how the quality of synthetic data generated by simulators impacts the performance of a DSS. When referring to data quality, we specifically mean the model's ability to accurately replicate the characteristics of the system under study in a realistic manner.

This work aims to explore the relationship between the realism of the simulator (its fidelity) and the predictive quality of a DSS that utilizes such a simulator. The study examines the significance of various model parameters and introduces seven quantifiable fidelity measures, extending earlier work [8]. Furthermore, the impact of these fidelities on a decision support model is analyzed in greater depth. Accurately assessing different aspects of simulator fidelity can support the development of future DSSs. Instead of repeatedly altering simulator parameters, generating synthetic data, and training the DSS, parameters might be fine-tuned to enhance a fidelity measure, accelerating the feedback loop for the developer and avoiding costly re-training of the DSS. The investigated methods are applied to a practical use case: a DSS focused on the scenario of liberating a village from occupying forces.

II. Related Work

The term "fidelity" originates from simulators and equipment that are used for training personnel [9]. In the context of DSSs for military operations, certain literature employs the term "fidelity", often in conjunction with the term "high" to describe the quality of a model [10], [11]. It's important to note that "high fidelity" does not necessarily mean highly detailed. While greater detail can enhance realism, it is not the sole determinant of fidelity. Fidelity is actually defined as the degree to which a virtual environment accurately replicates the real world [12]. Therefore, even an abstract model can exhibit high fidelity if it accurately portrays target behaviors. Consequently, the concept of fidelity can also be applied to rudimentary decision support applications and its generated synthetic data.

Intuitively, a DSS based on high-fidelity data seems desirable. Nonetheless, it has been shown that models defined as high-fidelity were not necessarily more effective than low-fidelity models [13], [14]. This assessment has been conducted specifically for models employed in the educational context, but abstract models have also shown their value in military applications. [15].

High-fidelity models typically entail substantial development costs, often necessitating the expertise of specialists [16]. Additionally, simulations with a high level of detail tend to be computationally expensive, resulting in lengthy execution times. Hence, to design a simulation system that demands minimal maintenance and expertise yet ensures high realism and accuracy, it is crucial to investigate how to achieve the optimal level of fidelity necessary for effective decision support. However, a persistent challenge in current literature remains how to precisely define and measure the fidelity of a simulator.

Recent efforts have been made to propose an objective framework for assessing the level of fidelity of a simulator. A common terminology often found in this area of research is "referent". A referent is an abstract concept which corresponds to the authoritative representation of the studied system [16]. In other words, the referent is the closest representation one has to the desired studied system.[1] By comparing the referent with the developed simulator and measuring the gap between them, one can define the fidelity of the simulator. The goal is to replicate the attributes and functionality of the referent as closely as possible to achieve a high-fidelity model. Thereby, some literature proposed more fine-grained fidelity characteristics, tailored to the requirements of the simulation. Different facets of fidelity as defined in [17] are listed below. Please note that these are defined in the context of a training simulator (e.g., a pilot in a simulated cockpit).

- ***Physical fidelity*** describes the degree of realism of the simulator in terms of somatosensation.
- ***Equipment fidelity*** describes the realism of the replicated equipment such as hardware and software.
- ***Visual-audio fidelity*** describes how the auditory and visual perception are stimulated.
- ***Task fidelity*** describes the degree of realism of the tasks and maneuvers to be executed.
- ***Functional fidelity*** describes the realism of interactions and behaviors of the elements within the simulation.
- ***Motion fidelity*** describes the degree of motion cues felt in the simulated environment.

Depending on the simulator's requirements, certain fidelity characteristics may be more developed than others. For instance, in a simulator designed for training submarine detection, the audio component of the simulation assumes greater importance [18].

Measuring the level of fidelity is a complex task, given that no clear definition exists. In most cases, the level of fidelity is not really assessed but only given [10], [11], [19]. Thereby, the classification often entails either *low*, *middle* or *high* fidelity. Furthermore, ordinal categories are often insufficiently descriptive and lack the finer granularity required for more detailed differentiation. Many articles give no details about the fidelity level or entail a highly subjective quantification [20].

When regarding a DSS, it is not the perceived realism by a human but rather the quality of the provided support to the human operator that is important. For this we look at the fidelity of the simulations, and investigate the correlation between the simulation fidelity and the quality of the decision support. We introduce seven novel quantifiable fidelity measures. When these fidelity facets prove valuable, they can be used during the development of future simulators to effectively create simulations that are usable for decision support.

Besides DSSs, the simulator fidelity plays a large role when trying to bridge the Sim2Real gap [21]. This gap indicates the difficulty in learning a behavior model in a simulator, and then executing the same behavior model in the real world, typically in a robot. Among other techniques, the general assumption is that a higher fidelity of the simulator makes the Sim2Real gap smaller [22]. Yet, the more complex the simulator, the easier it becomes to exploit it for AI systems [23]. One therefore may prefer to study the Sim2Real predictive capabilities rather than increasing the fidelity [24].

A DSS however does not directly interact with the environment. There is a human operator that can assess the advice and improves the course of action before effectuating it. For a DSS, we propose to call the inaccuracy of advise based on simulations, the *Sim2Advice* gap. Thus, this study aims to investigate the relationship between fidelity and the Sim2Advice gap in the context of military operations, specifically focusing on the liberation of a village. The objective is to understand how various fidelity characteristics of the simulators correlate with the performance of a DSS in accurately predicting the outcome of the mission and serving as reliable DSS.

The approach involves developing different simulators that replicate military exercises centered around the liberation of a village. Each simulator's fidelity is assessed. Subsequently, a classifier is trained independently on these simulators to predict the mission's outcome. It is hypothesized that this process will generate models with varying levels of predictive performance. By analyzing the correlation between simulator fidelity levels and the accuracy of AI predictions, the study

[1] If sufficient ground truth data is available, then the data itself may act as referent. However, for military applications such data is typically unavailable. Examples include dangerous flight maneuvers or the outcomes of a risky military course of action.

aims to identify the relation of fidelity measures and the effectiveness of AI-based DSSs in military contexts.

It must be noted that the study is a follow-up to [8], where the research explored how the quality of the synthetic data impacts the predictive accuracy of a simulator for scenario planning. It should be emphasized that in the previous study, the analysis focused on the outcomes of simulators based on specific fidelity scores. In contrast, the study presented in this article analyses the performance of a predictive model trained on these simulators in relation to their fidelity scores. Also, the focus in [8] was on one specific type of fidelity, the maneuver fidelity. In this report, further fidelity characteristics were developed to broaden the applicability of the end results.

III. Use Case

The Mobile Combat Training Centre (MCTC) [25], [26] was introduced in 2003 by the Dutch ministry of Defense and enables soldiers to practice combat in a realistic setting, but without using real ammunition. Lasers and sensors are utilized to simulate firing weapons. The system tracks the location of soldiers and vehicles, used ammunition, and health status. A variety of weapons (e.g., rifles, heavy machine guns, indirect fire), vehicles (e.g., Fennek, Boxer) and terrain (e.g., cross-country, urban) may be included in the exercise. Figure 1 shows a soldier training with the MCTC. Note the laser sensors on the helmet that register hits, and the laser on the rifle for firing at opposing forces.

Fig. 1. A soldier training in MCTC [26]

Table I provides a general overview of the exercise from which the data was obtained. Please note that soldiers wounded or killed in action may be revived in the data to allow the exercise to continue. This occurs particularly often with the red soldiers which are defending buildings that are being cleared. Additionally, as this is an exercise, the numbers may differ significantly from those in a real mission.

All the data that the systems generate are logged to provide a rough overview of the current state of the battlefield to the trainer and to support the after-action review. The logged MCTC data contains the location of soldiers and vehicles at

TABLE I
Overview of the MCTC exercise.

Duration of the scenario	20 h
Number of red soldiers	18
Number of blue soldiers	125
Total number of fire events	4.789
Surface area of terrain	0.54km^2
Type of terrain	Urban Terrain
Number of Buildings secured	26

regular intervals. Also, fire events, hit events, kill events, and vehicle associations are recorded (i.e., when a soldier enters or exits a vehicle). The consistency of the data is somewhat lacking in several aspects. Soldier locations are only provided every 15 seconds, and are snapped to a cell on a grid (with cell size of roughly 1m x 1m).[2] Sometimes soldiers move several grid cells at once, for example when driving quickly in a vehicle. It is also not always clear whether a soldier is inside or outside a building, as the wall of the building can run through the center of such a grid cell. Other limitations include that it is not always clear what soldiers are firing at, and boarding or disembarking of vehicles is noisy. These limitations are not a problem for gaining a rough overview of the state of the operation for which the data was intended, but do form an additional hurdle for training models [27].

In spite of the deficiencies, the recorded data is one of few examples in which actual mission data is available. It accurately reflects the behavior of the soldier, such as movement speed and maneuver choices, as the soldier had to make such choices in the exercise. One may still argue that the behavior of the soldier in a training setting is different to an actual mission. The continuous efforts of the trainers of the Dutch MoD aim to reduce this gap and to increase the learning effect to prepare soldiers for future missions.

An exercise was selected that took place in the Dutch training village Marnehuizen, which was entirely built to train Military Operations on Urban Terrain [28]. The Blue forces entered the village at the bridge in the north-east and were tasked to clear the village of enemy forces. A house-to-house battle was fought, which lasted 20 hours, until the last houses on the west side of the village were declared free of enemies. Figure 2 shows an excerpt of the village, in which simulated soldiers execute the same mission. The Blue forces occupy several buildings in the east, and move forward building by building, eliminating any red forces (west) they encounter. Even though only data from a single exercise was obtained, the maneuvering of all soldiers of the company in training during 20 hours of exercise is available. This is sufficient data to gain initial insights into measuring the fidelity of a simulator.

IV. Methodology

To investigate how the quality of the synthetic data influences the performance of decision support applications, the

[2] The logged coordinates have 13 digits after the decimal point, implying a high resolution. We assume that the loss of precision originates from the sensors used to measure the positions.

Fig. 2. Forces liberate the Marnehuizen training village.

referent-abstract model concept is used [20]. In other words, to assess the fidelity of an abstract model, a comparison to a referent model is performed. The referent model is considered the most accurate model available, closely mimicking the reality it represents. The referent model serves as a benchmark against which the abstract model's performance and fidelity can be evaluated.

This comparison provides insights into how closely the model approximates the real-world, and the use of real-world data in the referent model is encouraged. In order to facilitate a meaningful comparison between the referent and abstract models, it is essential that both models simulate the same scenario. Hence, for this research, the chosen scenario revolves around the liberation of a village from occupying forces as it is being performed in the MCTC data.

The data obtained from the MCTC serves to establish the initial conditions of the simulations, encompassing factors such as the deployment count of soldiers, their initial positions, as well as fired bullets, and individual health status. The MCTC data does not record human features, such as stress or fatigue. For real-world data there is no option to generate more data easily. When real-world data is used as referent, an underlying assumption of the presented approach is that behavior that resembles the few available data points is deemed realistic. An entirely different behavior, which may be tactically valid and realistic, cannot be validated by the shown approach when not present in the referent data.

In this study we also use the commercially available VR Forces package to act as a referent [29]. Such a referent is able to produce as much data as desired, and has the potential to create diverse continuation from the same state, leading to a cloud of possible behaviors, rather than the single one that was observed in the real-world data.

A. Agent-Based Simulation Model

For the simulation model, an Agent-Based Model (ABM) has been created (see Figure 3). It was not the aim of this study to make this model perfect in terms of tactical military movements, as we want to study the differences between the real world and the model. The ABM was implemented in Python, utilizing the MESA package [30]. To incorporate more advanced geo-spatial features, the geographic information system extension Mesa-Geo was used [31]. The terrain information was provided by OpenStreetMap.

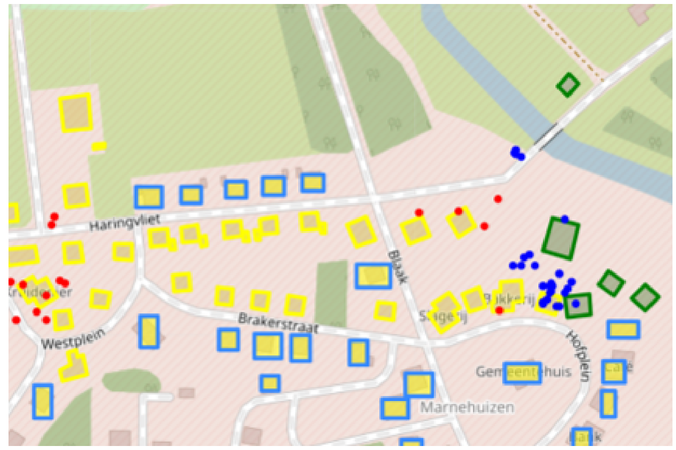

Fig. 3. Simulated forces are liberation Marnehuizen village.

The abstract ABM consists of an entity-level model, where two opposite forces are facing each other. The red forces adopt a random walk with minor fluctuations in the position, whereas the blue forces are moving towards the occupier. f The blue commander assigns a target zone for each blue unit to move towards. The target zone is defined by being between the centroid of the red forces and the centroid of the blue unit.[3] On the way to the targeted zone, the forces have to secure the closest building. The securing process functions similar to a progress bar, with each soldier's presence in the house incrementally increasing the value until it reaches 100%, at which point the house is considered secure. However, this only holds true if there are no enemy occupants in the house. If any unit (blue or red) has direct line of sight with the adversary, the unit opens fire.

The simulation ends when no enemy forces are left or all the buildings are cleared. During the simulation. random events may occur. An overview of these events are listed in Table II.

Moreover, the model encompasses three adjustable parameters denoted as sensory capacity, velocity and firing success, each of which exerts influence over the model's behavioral dynamics. The sensory capacity is defined as the distance, measured in meters, within which a soldier can perceive an enemy soldier. The velocity parameter quantifies the speed at which soldiers traverse their environment, expressed in kilometers per hour (km/h). The firing success metric is linked to the Monte-Carlo temperature and characterizes the likelihood that a firing attempt directed at the adversary results in success.

[3] In reality the blue commander might not know where the red forces are. This simplification is an easy way to ensure the termination of the scenario without hard-coding behavior.

TABLE II
THE RANDOMIZED EVENTS IN THE AGENT-BASED MODEL

Random events	Description
Soldier activation	The ordering in which each soldier takes actions each round is randomized.
Firing interaction	Whether a target has been hit is based on a certain probability threshold.
Soldier spots enemy	When having line of sight with multiple adversaries, an agent chooses each round a random adversary to fire at.
Define target zone	The target zone (zone to secure) for each squad is randomly defined by taking a randomly selected point along the interpolated line between the squad's centroid and the adversary's centroid.
Soldier movement	The movement of a soldier within the target zone is randomized.

B. VR-Forces Simulation Model

VR-Forces is a computer simulation that makes it possible to simulate entities in an environment [29]. The Marnehuizen terrain data is loaded into VR Forces. The movement and firing processes are already available in the simulation and can be used without any changes. How soldiers are moving through the terrain and how they are engaging each other is not available by default in the simulator. Normally the behavior is created by operators that interact with the simulator through a user interface. In our case the behavior of the units is determined algorithmically so that the simulator can run without using an operator. The plan contains tasks such as moving along a certain route with a chosen velocity, firing at a certain enemy and so on. The behavior plans contain also conditions on which the task should be executed (e.g., start firing when an enemy is detected, wait a certain time when entering a specific area).

In VR-Forces it is also possible to define new tasks by using the available tasks and conditions as building blocks. Such a definition is made inside a Lua script task. For the Marnehuizen scenario three new tasks are defined:

- **MoveToAttackPosition**. The blue soldier moves to a position that enables firing at an alive red soldier. The position is on a circle at a certain attack distance of the red soldier.
- **AttackClosestEnemy**. The soldier fires at an alive opponent soldier until either soldier is eliminated. When the opponent soldier is eliminated, the next opponent soldier within range is targeted, until there are no opposing soldiers left within range.
- **DefendAgainstClosestEnemy**. This is similar to **AttackClosestEnemy**. However, as defender the element of surprise is not in your favor. Therefore some parameters are changed to have a lower initial combat effectiveness.

Note that the success of a shot in the AttackClosestEnemy and DefendAgainstClosestEnemy is determined by a kill probability that is defined in VR-Forces, which depends on the weapon, target and ammunition, among others.

In the simulation each red soldier simply keeps repeating the DefendAgainstClosestEnemy action. Each blue soldier repeats executing the following plan until all red enemies are killed and all buildings are secured.

1) MoveToAttackPosition
2) AttackClosestEnemy
3) When inside an area of a building, wait for a certain time. This represents the required time for clearing a building.

As we want to run the scenario multiple times with different outcomes, additional random factors are included. Below, the included random factors are given:

- **MoveToAttackPosition**. The movement speed of a unit is the ordered speed reduced by factors such as terrain, weather, time of day. The ordered speed is randomly varied between simulations. Also, the attack position is randomized by altering the attack distance from the target.
- **AttackClosestEnemy** and **DefendAgainstClosestEnemy**. The firing order is varied.

In order to be able to compute the proposed fidelity measures, the following information is logged every 10 simulation seconds for each soldier:

- The simulation time.
- The soldier's identifier.
- The position of the soldier.
- A building identifier in case the soldier is inside a building.
- The status of the soldier (alive or dead)

When a fire event occurs, the following information is stored:

- The simulation time.
- The id and name of the soldier that is firing.
- The id and name of the soldier that is being fired at.

C. Fidelity Measurement

As stated in Section I, the fidelity can be fine-grained into more specific fidelity characteristics. In this research we propose, next to the maneuver fidelity as defined in [8], new fidelity aspects for computer generated forces, resulting in a total of seven fidelity measures. This section introduces all seven fidelity measures. In the definitions below, when mentioning "adequate fidelity measure," we refer to a fidelity measure capable of capturing a distinctive pattern between the fidelity score and the simulator's accuracy in predicting the correct mission time.

(1) The **Maneuver Fidelity** quantifies the disparity between the soldiers' paths in a simulation (abstract model) and those observed in a referent. In [8], a detailed explanation of the definition of maneuver fidelity is provided, including the implications of maneuver fidelity on the mission's simulated time in the abstract model. In this report, the maneuver fidelity is further developed by looking into the Dynamic Time Dependent Warping (DTDW) measure. The present study further investigates the formulated DTDW formula by examining the implications of the weights within the formula. In

other words, the investigation focuses on understanding which terms (distance or time between the points of two different paths) in the formula play a dominant role in adequately formulating maneuver fidelity. The results of this investigation are presented in Subsection V-A.

(2) The ***Fire Timing Fidelity*** captures the difference in timing of fire events between the model and the referent (i.e., the time between each successive firing at the same target). The fidelity score is obtained by taking the absolute mean difference of the distribution of fire timing in the model compared to the distribution of fire timing in the referent. In other words, for each repeated simulation with the same parameters, the fire timing fidelity was computed relative to the referent simulations. The mean was then taken for each simulation, and these results were averaged over all repetitions.

(3) The ***Firing Cover Fidelity*** captures the difference between model and referent in how effectively a soldier can use cover during engagements. To compute the cover of a soldier, the distance to the closest object (e.g., a wall of a house or a tree) is averaged with the distance to the closest squad member. The object is assumed to provide physical coverage, while the squad member may provide covering fire. Subsequently, the average coverage value of each firing soldier in the model is compared to the average coverage value of each firing soldier in the referent, the lower this difference, the better. This is averaged over each simulation, resulting in the firing cover fidelity measure.

(4) The ***Firing Distance Fidelity*** captures the absolute difference in the shooting range between the model and the referent. The mean shooting range is computed over all firing events within one simulation and averaged across simulations. The absolute difference between the mean shooting range of model and referent is the firing distance fidelity.

(5) The ***Firing Type Fidelity*** captures the difference in firing exchanges between soldier types in the model and the referent. In essence, the fidelity attempts to measure whether the fire exchanges between different soldier types (e.g., infantry, engineer, sniper) in the model make sense and correspond to what is observed in the referent. To compute the fidelity, the following equation was used

$$firing\ type\ fidelity = \frac{1}{n}\sum_{i=1}^{n} F_i^{-1} \tag{1}$$

where n is the number of simulation runs, and F_i^{-1} corresponds to the inverse F score of simulation i of the model. In this case a True Positive (TP) is recorded when, in both the referent and the model, a soldier of some type A fires at another soldier of type B. A False Positive (FP) corresponds to when this combination is not present in the referent but in the model. A False Negative (FN) corresponds to a combination that occurs in the referent, but not in the abstract model. The inverse of the F score was taken to ensure that as F increases, the gap between the abstract model and the referent decreases. Each simulation i of the model is compared to all simulations of the referent, resulting in n F scores that are inverted and averaged.

In this case, the specific soldier IDs, their location, and the timing of the firing are not important. Just the fact that a soldier of some type fires at a soldier of some other type is relevant to this fidelity.

(6) The ***Overall Firing Fidelity*** constitutes of multiple sub-fidelity criteria, namely the ***Firing Timing Fidelity, Firing Cover Fidelity, Firing Distance Fidelity, Firing Type Fidelity***. Due to the sub-fidelities being in different units and scales, each sub-fidelity score was transformed using a logarithmic function (i.e., applying $e^{sub-fidelity\ score}$) before averaging to calculate the overall firing fidelity. This approach aims to prevent scores that are an order of magnitude higher from disproportionately influencing the averaging of the fidelity scores.

(7) The ***Secured Zone Fidelity*** captures the differences of the zones secured in the model and the referent. A secured zone is defined as an area that has been cleared by the blue forces and is free from potential threats. In the Marnehuizen use case, each building is a zone. Thus, to compute the fidelity, the following equation is used

$$secured\ zone\ fidelity = \frac{1}{n}\sum_{i=1}^{n} (\mu(\Delta t_i)) * F_i^{-1} \tag{2}$$

where n is the number of simulation runs, μ denotes the average time difference in one simulation. F_i^{-1} corresponds to the inverse of the F score for simulation i. We count a TP when a zone has been captured in both the referent and the model, and a FP when a zone was captured in the model, but not in the referent. A FN secured zone corresponds to a secured zone in the referent model but not secured in the model. The timing difference $\mu(\Delta t_i)$ can only be computed for TPs. Each simulation i of the model is compared to all simulations of the referent, resulting in n F scores that are inverted and averaged.

D. Training a predictor for the DSS

In order to understand the relation between the fidelity of the model that generates synthetic data, and the performance of a DSS that has been trained on the synthetic data, we have implemented a rudimentary DSS that supports the user in estimating the remaining duration for liberating the village. A linear regression model is used to fit the synthetic data and predict the remaining mission time based on a given timestamp of the mission. The model is trained with one independent variable, the number of houses secured, and the dependent variable, the remaining mission time according to the simulations.

In order to be able to see the relation between simulators of a certain fidelity and the performance of the DSS, we varied the parameters of the ABM to obtain different behavior and thus different fidelity scores. The behavior of a simulator is heavily dependent on its parameter settings. The same simulator may show very different behavior when different parameters are given, and therefore has a different fidelity. From now on, when referring to *different* simulators, we also mean the same simulator (ABM or VR Forces) with different parameter settings. Simulators with similar fidelity scores were grouped into equal-sized buckets.

The regression model is then trained on data originating from different simulators from the same bucket. Subsequently, the regression model predicted the mission's outcome for unseen data originating from the referent, and the absolute error between the predicted time required to liberate the village and the actually required time according to the referent was computed.

V. Results & Discussion

This section presents the experiments, and start with tuning the DTDW measure for the maneuver fidelity in Subsection V-A. Thereafter Subsection V-B correlates all seven fidelity measures of a simulator to the performance of a DSS.

A. Tuning the DTDW measure for the Maneuver Fidelity

In order to see what the best weight α is between the distance term and the time term of the DTDW formula (see [8] for the formula, where a default weight of 0.5 is used), we vary the weight between 0 and 1 with 0.1 step sizes.[4] A high α value puts more weight on the spatial term, making it more important that the same route is followed, while a small α puts more weight on the timing of the movement. Figure 4 and Figure 5 show the relationship between the maneuver fidelity score and the simulator's generated mission time. Figure 4 computes the fidelity of the ABM with the actual soldiers movement as seen in the data, while Figure 5 compares the ABM with VR-Forces as referent.

It is observed that progressively shifting the importance from the timing term to the distance term makes the relationship between maneuver fidelity and simulated mission time more linear. It is interesting to point out that for $\alpha >= 0.6$ in Figure 4 and Figure 5, the linear patterns seem to indicate that if the fidelity score would further decrease (i.e., indicating a smaller gap between the soldier's path in the abstract model and the referent, and thus higher fidelity), the simulated mission's time would further increase. Thus, this would imply that the better the fidelity of the simulator, the more the simulated time would deviate from the actual 9 hours and become less accurate. These α values are therefore not usable for measuring the maneuver fidelity. The subfigures for $\alpha <= 0.4$ are much more intuitive, in which simulators that are generating either too short or too long mission times have lower fidelity.

This raises the question: What α produces the best correlation between a simulator's fidelity and the performance of a DSS? To answer this, we use the DSS as described in Subsection IV-D and compute the spearman correlation between a simulator's fidelity and the performance of the DSS. The results are presented in Figure 6 for both the comparison of the ABM to the MCTC data and to the VR-Forces.

[4] $\alpha = 0.5$ is omitted as it was already tested in earlier work.

One can see that with a lower α value, meaning more importance is placed on the timing of movements, the Spearman coefficient increases. For comparing ABM to MCTC data, the highest correlation value is achieved for $\alpha = 0$, but when comparing to VR-Forces, the best weight is $\alpha = 0.4$. It has to be noted that the VR-Forces results may show some noise, due to the fact that only 30 simulation runs could be made as the simulator is quite slow. We estimate that an α value of around 0.2 should work well across different referent models. For all further experiments shown in this section, $\alpha = 0.2$ is used for the maneuver fidelity.

B. Correlating a simulators fidelity to the performance of a DSS

In the experiments of this section we are interested in testing whether a higher fidelity of a simulator also implies better advice from a DSS, which in this case predicts the remaining mission time. As we have both the ABM and VR-Forces available, both can be used for producing training data of the DSS. We are also interested what the performance of the 7 defined fidelity measures are, and what measures perform best.

In the first experiment we focus on training the DSS based VR-Forces data. 30 simulations have been run in VR forces, and these data points are fed to the linear regression model for training. The resulting fidelity numbers for each of the 7 measures are reported in Table III. As VR-Forces is too slow to generate sufficient data for different parameter settings, we can only report the fidelities of a single setting.

TABLE III
The fidelity values of VR Forces compared to the MCTC data as referent.

Fidelity Measurements	Avg.	SD
Maneuver Fidelity	0.072	0.006
Firing Timing Fidelity	0.157	0.134
Firing Cover Fidelity	34.258	7.327
Firing Distance Fidelity	32.842	1.357
Firing Type Fidelity	1.965	0.726
Overall Firing Fidelity	2.869	0.747
Secured Zone Fidelity	2.088	0.327

In order to interpret the numbers reported in Table III, one has to be aware of the fidelity definitions. Not all fidelities values can be directly understood, and we will focus our explanation to the easier ones. We see that firing distance fidelity is 32.842, indicating that the difference in firing distance is on average less than 33 meters of between VR-Forces and the MCTC data. This is rather a small number, and can easily be explained as when modeling the scenario in VR-Forces, the actual MCTC was used as inspiration.

The firing type fidelity is 1.965 and the secured zone fidelity is 2.088. We experience that it is quite hard to judge whether the fidelity of VR-Forces can be judged as *high* or *low* simply by looking at these numbers. We conjecture that this becomes easier once the fidelity metrics become established and reference experiments are available, so that simulators can be compared to one another based on their fidelity scores.

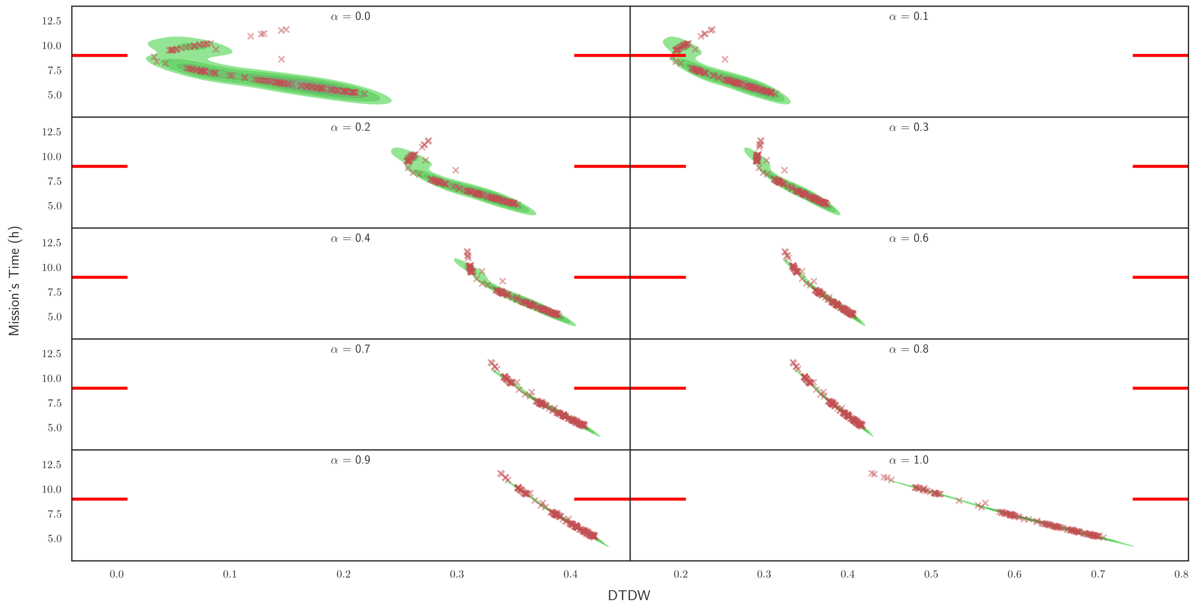

Fig. 4. Investigating the impact of the temporal term and spatial term of the DTDW formula on the relation between the fidelity score and the ABM's accuracy in simulating the correct mission time. The fidelity was computed using the ABM as abstract model and the MCTC data as referent. A high α gives more weight to the spatial term, while a low α gives more weight to the temporal term. A red cross corresponds to the average of one distinct simulator with its unique parameter combination as described in Section Subsection IV-A. The underlying green blob corresponds to the kernel density estimation of the red crosses. The horizontal lines correspond to the actual mission time of 9 hours.

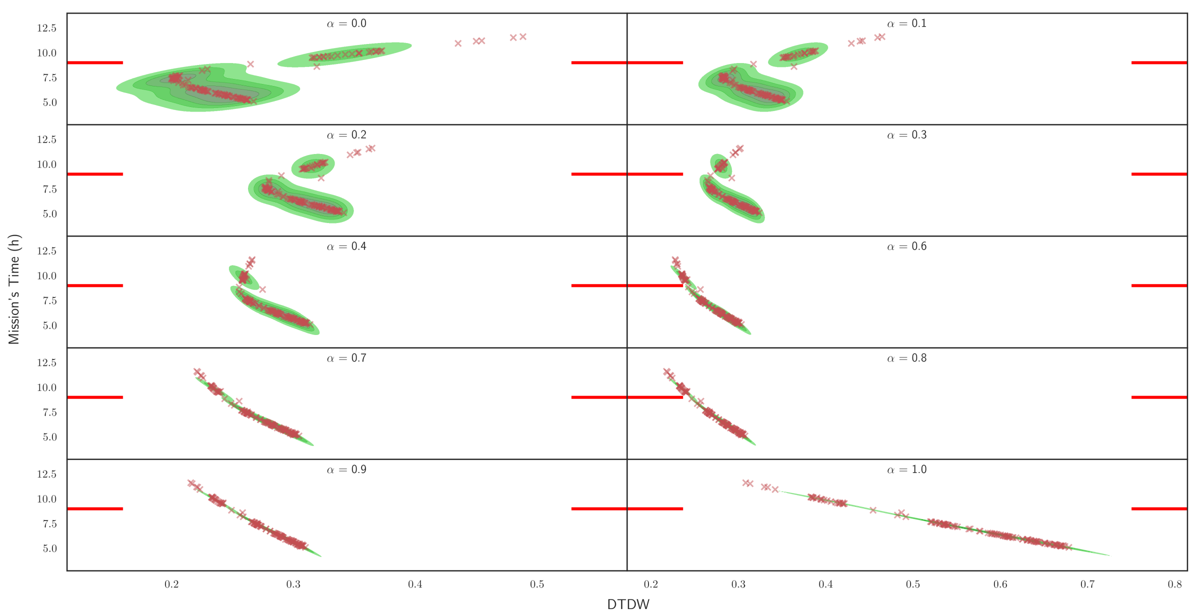

Fig. 5. Investigating the impact of the temporal term and spatial term of the DTDW formula on the relation between the fidelity score and the ABM's accuracy in simulating the correct mission time. The fidelity was computed using the ABM as abstract model and the VR Forces data as referent. A high α gives more weight to the spatial term, while a low α gives more weight to the temporal term. A red cross corresponds to the average of one distinct simulator with its unique parameter combination as described in Section Subsection IV-A. The underlying green blob corresponds to the kernel density estimation of the red crosses. The horizontal lines correspond the the actual mission time of 9 hours.

In the last experiment, we are using the ABM which allows us to generate sufficient data for different parameters, and thus different fidelity values. We show how the fidelity measure correlates to the performance of the DSS that is trained on the data of the respective fidelity number, by computing the well-known spearman coefficient. The performance of the DSS is defined as the absolute difference between the predicted remaining mission time, and the actual remaining mission time. We use different states of the actual MCTC data at different times as data points for measuring the performance.

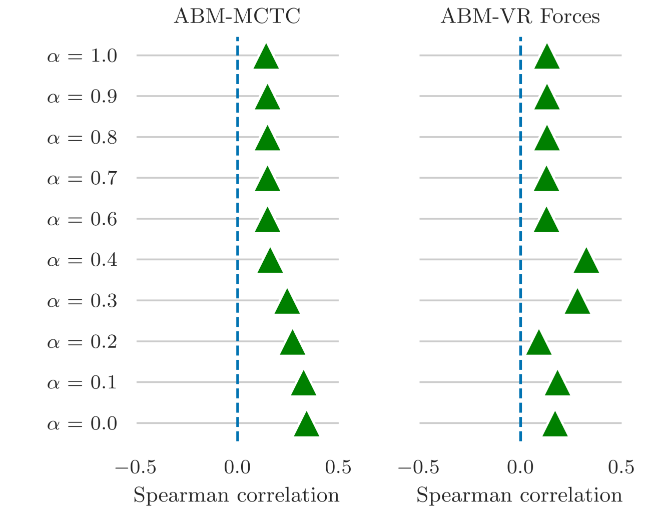

Fig. 6. The Spearman coefficient for different α weights of the DTDW formula and the absolute error of the model in predicting the remaining mission time given a specific timestamp within the simulation.

In a second experiment we compare the ABM to VR-Forces as referent, with the same conceptional setup. In this second test we are however using all 30 VR-forces simulations as referent, compared to only the single MCTC data point. The results for each of the 7 fidelity measures are presented in Figure 7.

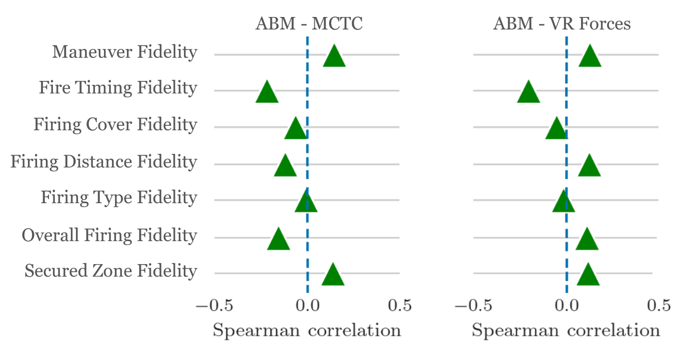

Fig. 7. The Spearman coefficient for different fidelities and the absolute error of the model in predicting the remaining mission time given a specific timestamp within the simulation. The Spearman coefficient is computed for the fidelities determined using the ABM - MCTC data, ABM - VR Forces.

We observe that the building secured and maneuver fidelity have positive spearman correlations for both the ABM-MCTC setting and the ABM-VR Forces. This indicates that these fidelity measures work across different referents. Although the spearman correlation is still quite low, the reader should be aware that simulations can vary greatly after certain randomized events occur. A spearman value of 1 is therefore not realistic. Whether the correlation is sufficient to aid a developer with the creation of simulators remains to be tested.

Furthermore, the firing distance fidelity and the overall firing fidelity have positive correlations with VR forces as referent, but not with ABM as referent. These fidelity measures are promising, but we believe more work is required to make these robust across simulators.

The firing type-, fire timing- and firing cover fidelity show negative spearman correlations on both simulators. This indicates that these fidelity measures are currently not usable.

VI. Conclusion

This work aimed to explore the relationship between the realism of the simulator (its fidelity) and the predictive quality of a DSS that utilizes such a simulator. For this, a military scenario of clearing the village of Marnehuizen was used, for which actual data is available. An agent-based model and VR Forces were configured to simulate this mission. The study introduced 7 quantifiable fidelity measures that are relevant for the scenario, extending earlier work [8]. We fine-tuned the maneuver fidelity and tested the performance of all 7 measures in various combinations of agent-based simulation, VR Forces and actual data.

The results of fine-tuning the DTDW formula of the maneuver fidelity showed that an α value of 0.2 worked best, which places most weight on the timing component. This setting showed consistent results when comparing the simulated mission to actual data, and is correlated well to the classifiers performance of the DSS.

We also tested how the 7 fidelity measures correlate to the performance of a DSS that is trained on the data of the respective fidelity number, by computing the well-known spearman coefficient. The performance of the DSS is defined as the absolute difference between the predicted remaining mission time, and the actual remaining mission time. We observed that the building secured and maneuver fidelity have positive correlations and are robust across referents. The firing distance fidelity and the overall firing fidelity are promising, but we believe more work is required to make these robust across simulators. The firing type fidelity, fire timing fidelity and firing cover fidelity are not yet suitable for use.

We believe that the approach of creating quantifiable fidelity measures for accurately assessing different aspects of simulator fidelity can support the development of future DSSs. Instead of repeatedly altering simulator parameters, generating synthetic data, and training the DSS, parameters might be fine-tuned to enhance a fidelity measure, accelerating the feedback loop for the developer and avoiding costly re-training of the DSS. Proving the added value during the development of a simulator is subject of future research.

We furthermore would like to investigate whether the firing type fidelity may be improved by counting the number of exchanges between soldier types. Currently, the measure counts a true positive when a matching firing exchange happens in both simulator and referent. For example, if in the simulator an infantry soldier fired once at an engineer, while in the referent this happened 100 times, this difference is currently not taken into account. We would like to redesign the formula to address this issue and test the performance of this variant.

In spite of the promising results for the investigated scenario, it is still unclear wether the results generalize to different scenarios. The primary aim was to lay the groundwork for investigating the impact of fidelity on decision support applications within the military domain. We hope this initial study will inspire further research to explore this topic across different scenarios.

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